ModelIt!

Student Activity Pack

How Do Fossils Form?

Clues from Ancient Life

Grade 3 | NGSS 3-LS4-1 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever seen a fossil? What did it look like and where was it?

2 How do you think the bones of a dinosaur that lived millions of years ago can still exist today?

3 Draw what you think happens to an animal's bones after it dies and gets buried underground.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Burial Depth | Pressure | Fossilization

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Shallow Burial

Burial Depth set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Deep Burial

Burial Depth set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Medium Burial

Burial Depth set to 50%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Mineral Content — How many dissolved minerals are in the groundwater — more minerals means faster replacement of original material

Time — How many millions of years the remains have been buried — more time allows more complete fossilization

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do the bones and shells of ancient animals turn into rock fossils that last millions of years?
Use evidence from your simulation to support your answers.

1 Why do most animals that die NOT become fossils, and what makes some animals more likely to fossilize?

2 How did your model help you understand why fossils take millions of years to form?

8 STEM Challenge: Create a Fossil Time Capsule

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YOUR ENGINEERING MISSION

Design a model that shows how fossils form step by step, creating a layered fossil time capsule that demonstrates the burial and fossilization process.

THE SCENARIO

A natural history museum wants a hands-on exhibit that shows visitors how fossils form. Your team must create a layered model that demonstrates burial, pressure, and fossilization step by step. Use what you learned about the fossilization process!

Engineering Design Thinking

1. WHAT WILL YOU USE FOR THE ANIMAL REMAINS — A SHELL, A PLASTIC BONE, OR A LEAF?

2. WHAT MATERIALS WILL REPRESENT SEDIMENT LAYERS — SAND, CLAY, PLASTER?

3. HOW WILL YOU STACK THE LAYERS TO SHOW TIME PASSING?

4. HOW WILL YOU LABEL YOUR MODEL SO MUSEUM VISITORS CAN UNDERSTAND EACH STEP?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Paleontologists travel the world digging up fossils and studying ancient life to understand the history of our planet. They earn $50,000-$95,000/year.